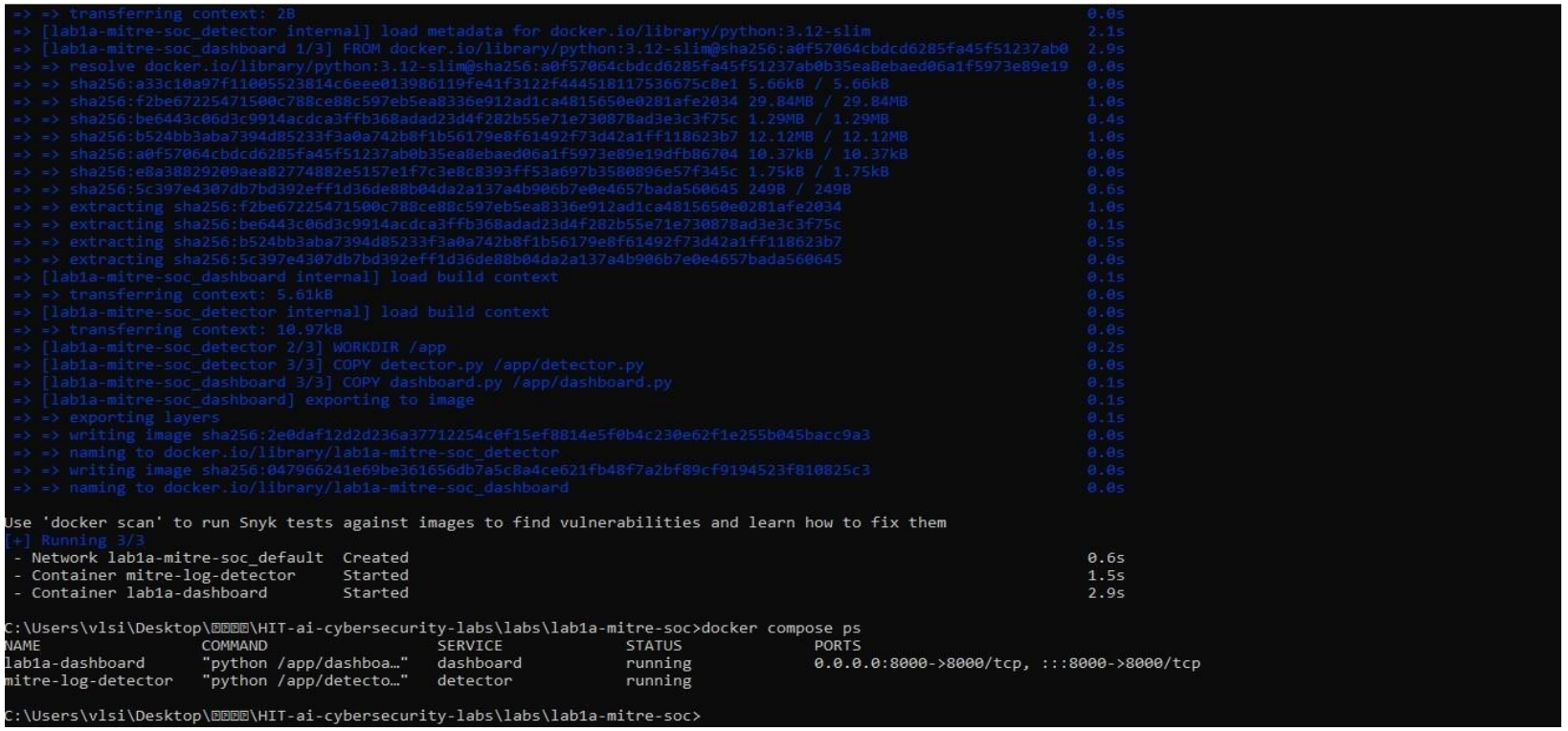
1 Docker environment verification

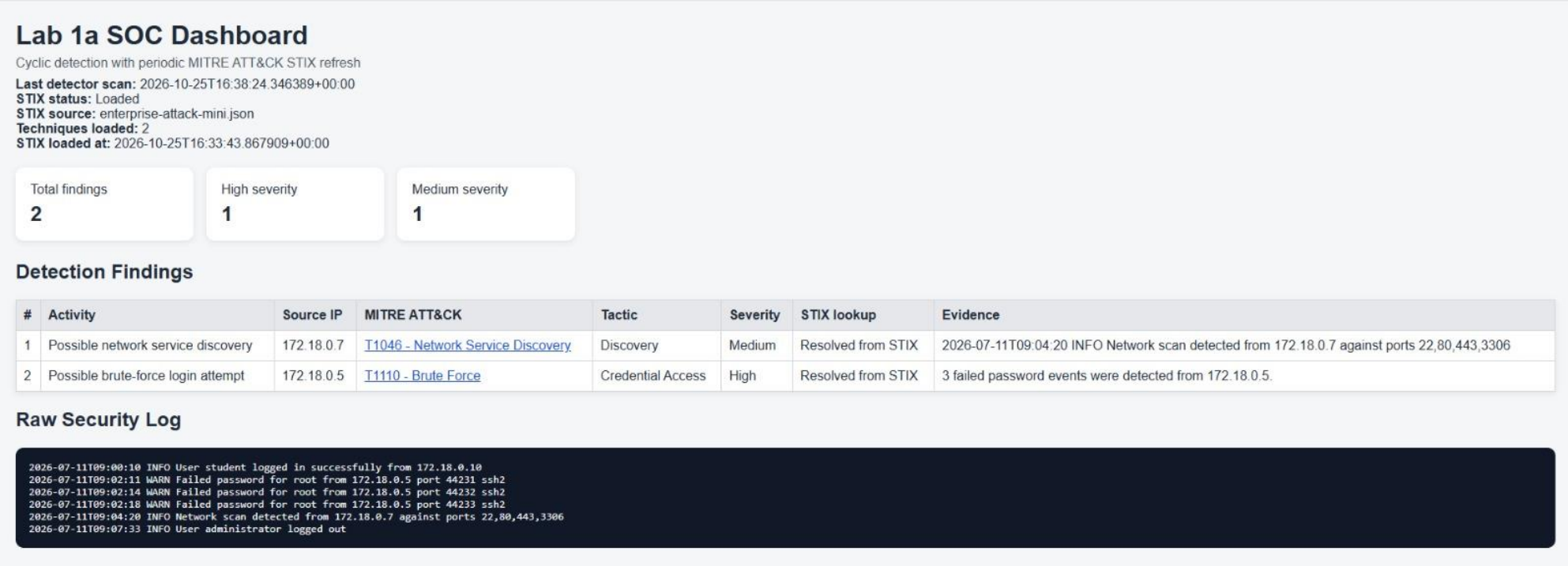
This laboratory examines rule-based detection of security log events and their mapping to MITRE ATT&CK techniques. The Docker images were built and the detector and dashboard services were started.



The docker compose ps output shows both services as running. The dashboard is mapped to port 8000. This verifies that the environment was started. The detector is deterministic and rule-based. We verified Docker using docker version and validated the Compose configuration with docker compose config. After startup, we checked docker compose ps and the detector and dashboard service logs to confirm that both services were running and the detector was generating output.

2 Clean baseline

The original log contains six records: one successful login, three failed-password events, one network-scan notice and one logout. The dashboard shows two findings, as expected.



Lab 1a SOC Dashboard

Cyclic detection with periodic MITRE ATT&CK STIX refresh

Last detector scan: 2026-10-25T16:38:24.346389+00:00
STIX status: Loaded
STIX source: enterprise-attack-mini.json
Techniques loaded: 2
STIX loaded at: 2026-10-25T16:33:43.867909+00:00

Total findings	High severity	Medium severity
2	1	1

Detection Findings

#	Activity	Source IP	MITRE ATT&CK	Tactic	Severity	STIX lookup	Evidence
1	Possible network service discovery	172.18.0.7	T1046 - Network Service Discovery	Discovery	Medium	Resolved from STIX	2026-07-11T09:04:20 INFO Network scan detected from 172.18.0.7 against ports 22,80,443,3306
2	Possible brute-force login attempt	172.18.0.5	T1110 - Brute Force	Credential Access	High	Resolved from STIX	3 failed password events were detected from 172.18.0.5.

Raw Security Log

```
2026-07-11T09:00:10 INFO User student logged in successfully from 172.18.0.10
2026-07-11T09:02:11 WARN Failed password for root from 172.18.0.5 port 44231 ssh2
2026-07-11T09:02:14 WARN Failed password for root from 172.18.0.5 port 44232 ssh2
2026-07-11T09:02:18 WARN Failed password for root from 172.18.0.5 port 44233 ssh2
2026-07-11T09:04:20 INFO Network scan detected from 172.18.0.7 against ports 22,80,443,3306
2026-07-11T09:07:33 INFO User administrator logged out
```

T1110: three failed-password events from 172.18.0.5 produce a High-severity finding under Credential Access. The scan notice from 172.18.0.7 produces a Medium-severity T1046 finding under Discovery.

3 Detector output and MITRE metadata

The findings.json output includes evidence, detection rules, source IPs, severity and MITRE mappings. The screenshot shows three failed attempts from the original source IP, along with tactic names and technique links.


```
    {
      "finding": "Possible brute-force login attempt",
      "source_ip": "172.18.0.5",
      "failed_attempts": 3,
      "candidate_mitre_technique": "T1110",
      "severity": "High",
      "evidence": "3 failed password events were detected from 172.18.0.5.",
      "detection_rule": "At least 3 failed password events from the same source IP.",
      "mitre_technique": "T1110",
      "mitre_name": "Brute Force",
      "mitre_tactics": [
        "Credential Access"
      ],
      "tactic": "Credential Access",
      "mitre_url": "https://attack.mitre.org/techniques/T1110/",
      "mitre_stix_id": "attack-pattern--lab1a-t1110",
      "mitre_description": "Adversaries may use brute force techniques to gain access to accounts when passwords are unknown or when password hashes are obtained.",
      "mitre_created": "2026-07-11T00:00:00.000Z",
      "mitre_modified": "2026-07-11T00:00:00.000Z",
      "mitre_version": "1.7",
      "mitre_lookup_status": "Resolved from STIX"
    }
  ]
}
```


Resolved from STIX means that technique metadata was found in the local knowledge file. It verifies metadata enrichment, not that an attack occurred. This screenshot shows the enriched detector output rather than the knowledge file itself.

We inspected mitre/enterprise-attack-mini.json and its STIX attack-pattern objects. We checked the external technique references and tactic phases for T1110 and T1046 against the enriched findings.json output; the technique names and tactics matched.

4 Evidence and interpretation

Evidence	Detection rule	Technique	Tactic	Uncertainty
Three Failed password records from 172.18.0.5.	At least three failed-password events from the same source IP. No time window.	T1110 Brute Force	Credential Access	Does not prove successful access, attacker identity or malicious intent.
An explicit scan notice from 172.18.0.7 against ports 22,80,443,3306.	Matches the text network scan detected in a log record.	T1046 Network Service Discovery	Discovery	Relies on an explicit message; does not independently analyze packets or prove malicious scanning.

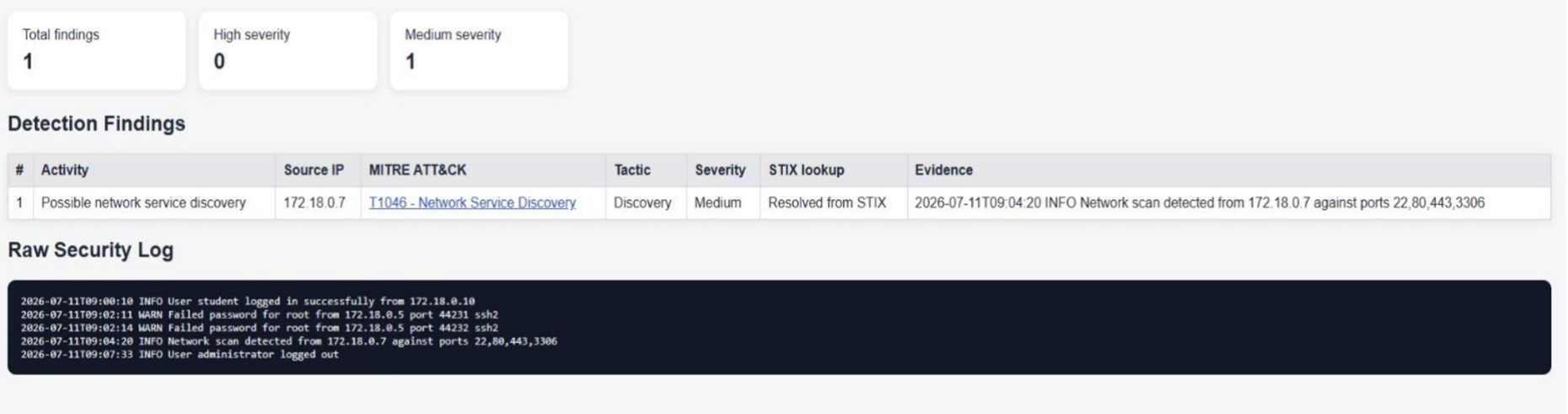
The lab describes FAILED_PASSWORD_THRESHOLD as 3 and a detector rescan interval of 5 seconds. Failed attempts are grouped by source IP. A rule match identifies suspicious behavior requiring further investigation; it does not establish a compromise.

We inspected detector.py: FAILED_PASSWORD_THRESHOLD = 3 controls the per-IP failure threshold, while SCAN_INTERVAL_SECONDS = 5 controls rescanning.
The authentication and scan-message rules assign candidate IDs T1110 and T1046, respectively.

Before editing, we backed up security.log as security.log.bak and stopped the simulator for C1-C4. After each saved change, we waited at least 5 seconds, refreshed the dashboard and compared it with findings.json. After observing the simulator, we stopped it before finalizing the evidence.

C1 Below the detection threshold

Log change: The third failed-password record from 172.18.0.5 at 09:02:18 was removed, leaving two failures from this source.



Prediction: T1110 disappears; the original T1046 remains.

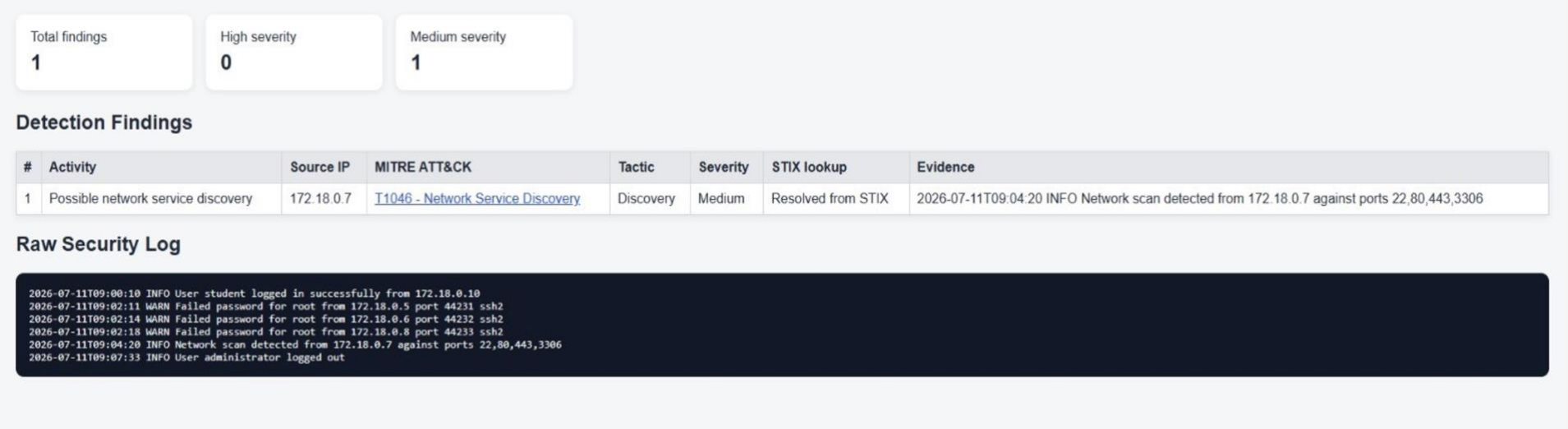
Observed result: One finding: zero High and one Medium. The result matches the prediction.

Explanation: Two failures are below the threshold of three. The scan notice was unchanged.

JSON verification: findings.json contained 1 finding: T1046, source_ip 172.18.0.7, Medium severity. No T1110 finding remained. The JSON count, source IP, technique and evidence matched the dashboard and the two-failure log state.

C2 Failures from three different IPs

Log change: Three failed-password records were restored. Their source IPs at 09:02:11, 09:02:14 and 09:02:18 were set to 172.18.0.5, 172.18.0.6 and 172.18.0.8, respectively.



Total findings	High severity	Medium severity
1	0	1

Detection Findings

#	Activity	Source IP	MITRE ATT&CK	Tactic	Severity	STIX lookup	Evidence
1	Possible network service discovery	172.18.0.7	T1046 - Network Service Discovery	Discovery	Medium	Resolved from STIX	2026-07-11T09:04:20 INFO Network scan detected from 172.18.0.7 against ports 22,80,443,3306

Raw Security Log

```
2026-07-11T09:00:10 INFO User student logged in successfully from 172.18.0.10
2026-07-11T09:02:11 WARN Failed password for root from 172.18.0.5 port 44231 ssh2
2026-07-11T09:02:14 WARN Failed password for root from 172.18.0.6 port 44232 ssh2
2026-07-11T09:02:18 WARN Failed password for root from 172.18.0.8 port 44233 ssh2
2026-07-11T09:04:20 INFO Network scan detected from 172.18.0.7 against ports 22,80,443,3306
2026-07-11T09:07:33 INFO User administrator logged out
```

Prediction: No T1110 finding; the original T1046 remains.

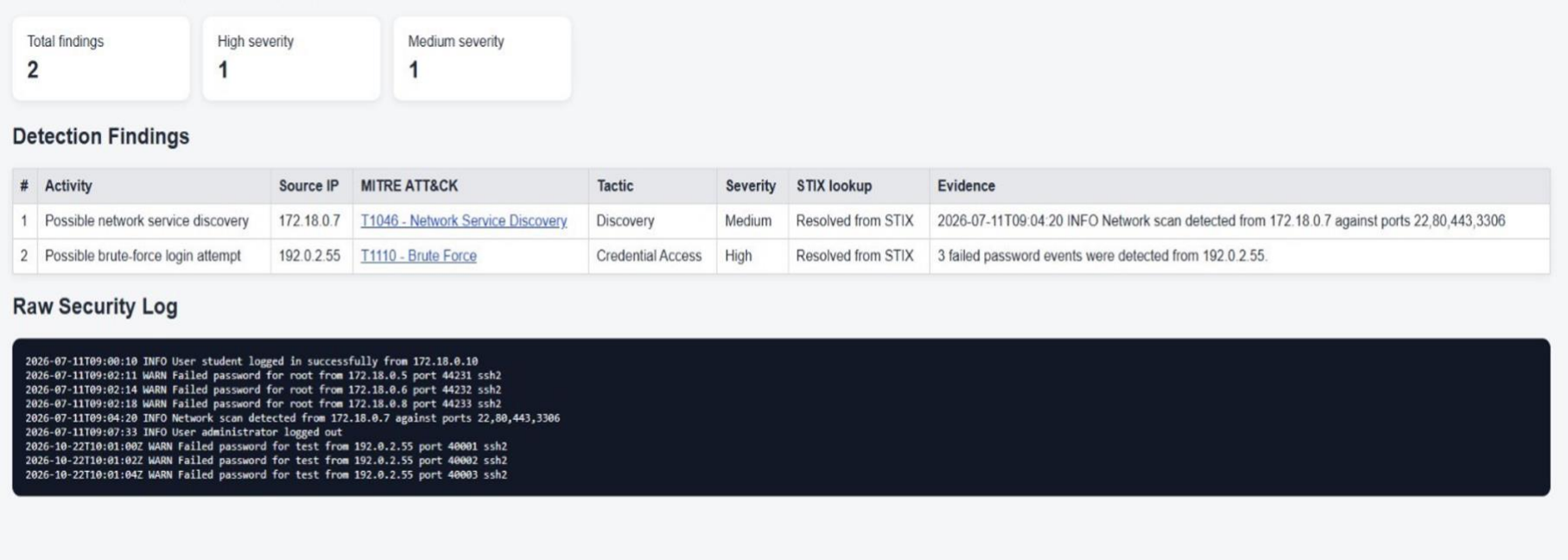
Observed result: One Medium-severity finding and zero High findings. The result matches the prediction.

Explanation: The detector groups failures by source IP. Each IP has only one failure, so none reaches the threshold.

JSON verification: findings.json contained 1 finding: T1046, source_ip 172.18.0.7, Medium severity, with no T1110. The result matched the dashboard: each authentication source IP had only one failed-password event.

C3 Threshold crossing from one new IP

Log change: After C2, three Failed password records for user test from 192.0.2.55 were appended, dated 2026-10-22 at 10:01:00Z, 10:01:02Z and 10:01:04Z, using ports 40001, 40002 and 40003.



Total findings	High severity	Medium severity
2	1	1

Detection Findings

#	Activity	Source IP	MITRE ATT&CK	Tactic	Severity	STIX lookup	Evidence
1	Possible network service discovery	172.18.0.7	T1046 - Network Service Discovery	Discovery	Medium	Resolved from STIX	2026-07-11T09:04:20 INFO Network scan detected from 172.18.0.7 against ports 22,80,443,3306
2	Possible brute-force login attempt	192.0.2.55	T1110 - Brute Force	Credential Access	High	Resolved from STIX	3 failed password events were detected from 192.0.2.55.

Raw Security Log

```
2026-07-11T09:00:10 INFO User student logged in successfully from 172.18.0.10
2026-07-11T09:02:11 WARN Failed password for root from 172.18.0.5 port 44231 ssh2
2026-07-11T09:02:14 WARN Failed password for root from 172.18.0.6 port 44232 ssh2
2026-07-11T09:02:18 WARN Failed password for root from 172.18.0.8 port 44233 ssh2
2026-07-11T09:04:20 INFO Network scan detected from 172.18.0.7 against ports 22,80,443,3306
2026-07-11T09:07:33 INFO User administrator logged out
2026-10-22T10:01:00Z WARN Failed password for test from 192.0.2.55 port 40001 ssh2
2026-10-22T10:01:02Z WARN Failed password for test from 192.0.2.55 port 40002 ssh2
2026-10-22T10:01:04Z WARN Failed password for test from 192.0.2.55 port 40003 ssh2
```

Prediction: A new T1110 for 192.0.2.55 appears alongside the original T1046.

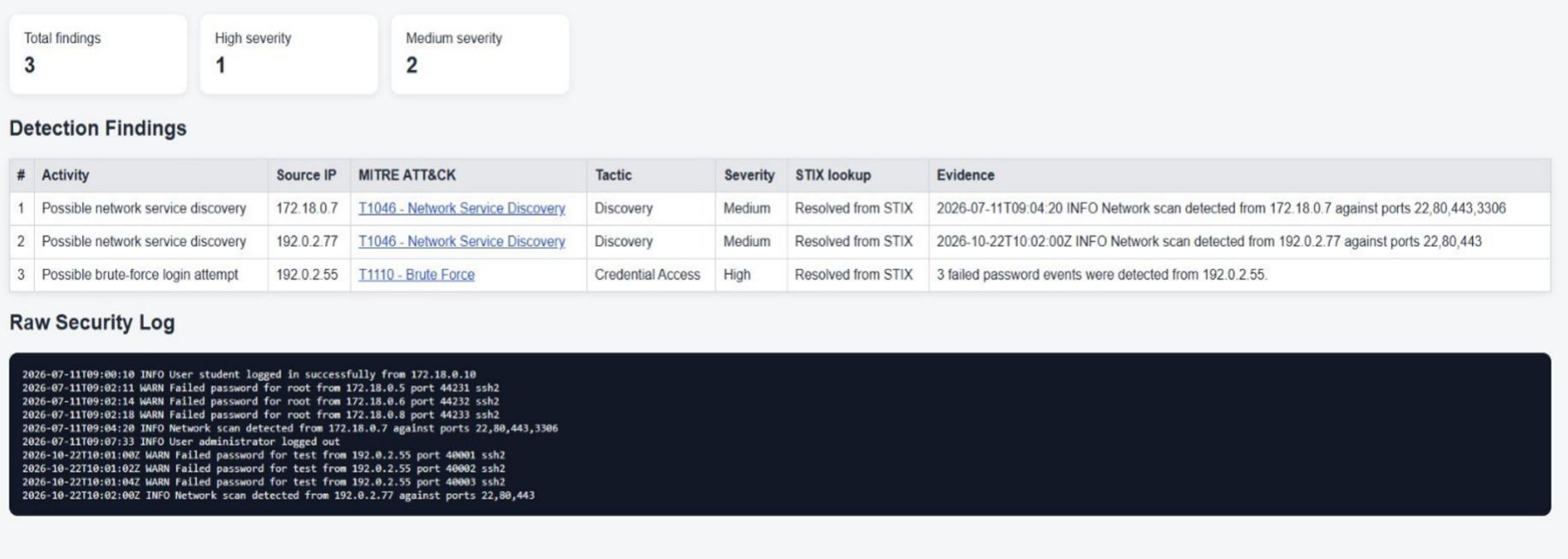
Observed result: Two findings: one High and one Medium. The result matches the prediction.

Explanation: Three failures from the same IP reach the threshold. The distributed failures from C2 still do not create a finding.

JSON verification: findings.json contained 2 findings: T1046 / 172.18.0.7 / Medium and T1110 / 192.0.2.55 / High. The count, source IPs and techniques matched the dashboard; the T1110 evidence recorded three failed-password events.

C4 Add a second network-scan notice

Log change: After C3, a record was appended dated 2026-10-22 at 10:02:00Z: Network scan detected from 192.0.2.77 against ports 22,80,443.



Total findings
3

High severity
1

Medium severity
2

Detection Findings

#	Activity	Source IP	MITRE ATT&CK	Tactic	Severity	STIX lookup	Evidence
1	Possible network service discovery	172.18.0.7	T1046 - Network Service Discovery	Discovery	Medium	Resolved from STIX	2026-07-11T09:04:20 INFO Network scan detected from 172.18.0.7 against ports 22,80,443,3306
2	Possible network service discovery	192.0.2.77	T1046 - Network Service Discovery	Discovery	Medium	Resolved from STIX	2026-10-22T10:02:00Z INFO Network scan detected from 192.0.2.77 against ports 22,80,443
3	Possible brute-force login attempt	192.0.2.55	T1110 - Brute Force	Credential Access	High	Resolved from STIX	3 failed password events were detected from 192.0.2.55.

Raw Security Log

```
2026-07-11T09:00:10 INFO User student logged in successfully from 172.18.0.10
2026-07-11T09:02:11 WARN Failed password for root from 172.18.0.5 port 44231 ssh2
2026-07-11T09:02:14 WARN Failed password for root from 172.18.0.6 port 44232 ssh2
2026-07-11T09:02:18 WARN Failed password for root from 172.18.0.8 port 44233 ssh2
2026-07-11T09:04:20 INFO Network scan detected from 172.18.0.7 against ports 22,80,443,3306
2026-07-11T09:07:33 INFO User administrator logged out
2026-10-22T10:01:00Z WARN Failed password for test from 192.0.2.55 port 40001 ssh2
2026-10-22T10:01:02Z WARN Failed password for test from 192.0.2.55 port 40002 ssh2
2026-10-22T10:01:04Z WARN Failed password for test from 192.0.2.55 port 40003 ssh2
2026-10-22T10:02:00Z INFO Network scan detected from 192.0.2.77 against ports 22,80,443
```

Prediction: An additional T1046 appears; the total rises from two to three.

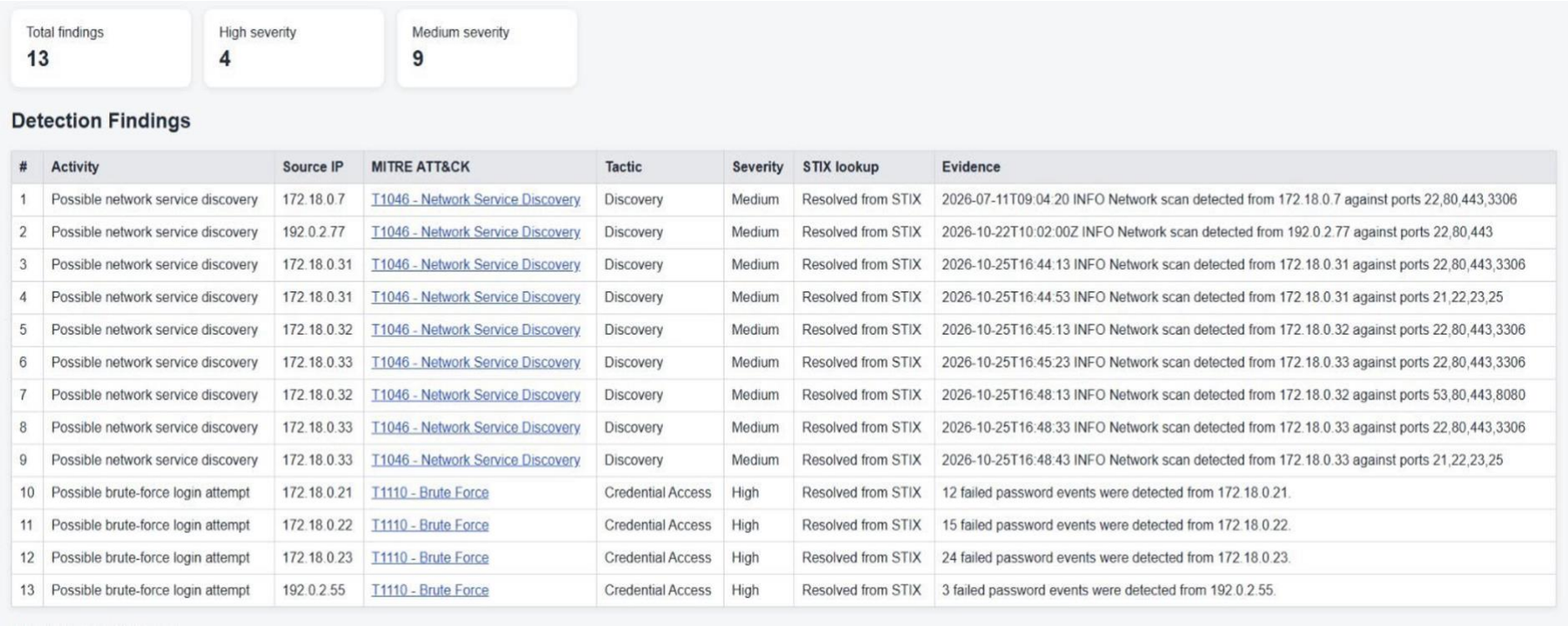
Observed result: Three findings: one High and two Medium. The result matches the prediction.

Explanation: The explicit scan text triggers the rule. The two previous findings remain because their supporting records are still present.

JSON verification: findings.json contained 3 findings: T1046 / 172.18.0.7 / Medium, T1110 / 192.0.2.55 / High and T1046 / 192.0.2.77 / Medium. The count, source IPs, techniques and supporting evidence matched the dashboard.

9 Simulator observations

The screenshot taken after the simulator was started shows 13 findings: four High and nine Medium. Compared with the three findings at the end of C4, ten additional findings are present.



Total findings	High severity	Medium severity
13	4	9

Detection Findings

#	Activity	Source IP	MITRE ATT&CK	Tactic	Severity	STIX lookup	Evidence
1	Possible network service discovery	172.18.0.7	T1046 - Network Service Discovery	Discovery	Medium	Resolved from STIX	2026-07-11T09:04:20 INFO Network scan detected from 172.18.0.7 against ports 22,80,443,3306
2	Possible network service discovery	192.0.2.77	T1046 - Network Service Discovery	Discovery	Medium	Resolved from STIX	2026-10-22T10:02:00Z INFO Network scan detected from 192.0.2.77 against ports 22,80,443
3	Possible network service discovery	172.18.0.31	T1046 - Network Service Discovery	Discovery	Medium	Resolved from STIX	2026-10-25T16:44:13 INFO Network scan detected from 172.18.0.31 against ports 22,80,443,3306
4	Possible network service discovery	172.18.0.31	T1046 - Network Service Discovery	Discovery	Medium	Resolved from STIX	2026-10-25T16:44:53 INFO Network scan detected from 172.18.0.31 against ports 21,22,23,25
5	Possible network service discovery	172.18.0.32	T1046 - Network Service Discovery	Discovery	Medium	Resolved from STIX	2026-10-25T16:45:13 INFO Network scan detected from 172.18.0.32 against ports 22,80,443,3306
6	Possible network service discovery	172.18.0.33	T1046 - Network Service Discovery	Discovery	Medium	Resolved from STIX	2026-10-25T16:45:23 INFO Network scan detected from 172.18.0.33 against ports 22,80,443,3306
7	Possible network service discovery	172.18.0.32	T1046 - Network Service Discovery	Discovery	Medium	Resolved from STIX	2026-10-25T16:48:13 INFO Network scan detected from 172.18.0.32 against ports 53,80,443,8080
8	Possible network service discovery	172.18.0.33	T1046 - Network Service Discovery	Discovery	Medium	Resolved from STIX	2026-10-25T16:48:33 INFO Network scan detected from 172.18.0.33 against ports 22,80,443,3306
9	Possible network service discovery	172.18.0.33	T1046 - Network Service Discovery	Discovery	Medium	Resolved from STIX	2026-10-25T16:48:43 INFO Network scan detected from 172.18.0.33 against ports 21,22,23,25
10	Possible brute-force login attempt	172.18.0.21	T1110 - Brute Force	Credential Access	High	Resolved from STIX	12 failed password events were detected from 172.18.0.21.
11	Possible brute-force login attempt	172.18.0.22	T1110 - Brute Force	Credential Access	High	Resolved from STIX	15 failed password events were detected from 172.18.0.22.
12	Possible brute-force login attempt	172.18.0.23	T1110 - Brute Force	Credential Access	High	Resolved from STIX	24 failed password events were detected from 172.18.0.23.
13	Possible brute-force login attempt	192.0.2.55	T1110 - Brute Force	Credential Access	High	Resolved from STIX	3 failed password events were detected from 192.0.2.55.

New source IPs 172.18.0.21, 172.18.0.22 and 172.18.0.23 generated T1110 findings with 12, 15 and 24 failures, respectively. Scan notices from 172.18.0.31, 172.18.0.32 and 172.18.0.33 generated T1046 findings.

The simulator appends events over time while the detector rescans the accumulated log. Counts may therefore change and depend on runtime and generated events, making this less reproducible than manual experiments.

10 SOC narrative and conclusions

The baseline log contained three failed-password events from 172.18.0.5 and a network-scan notice from 172.18.0.7. The detector produced two findings: High-severity T1110 and Medium-severity T1046. Reducing failures to two removed T1110. Distributing three failures across different IPs did not trigger it. Adding three failures from one new IP generated T1110, while appending another scan notice generated an additional T1046. After simulator activity, the dashboard displayed thirteen findings, including new source IPs. These observations demonstrate sensitivity to thresholds, source-IP grouping and explicit log text. A possible interpretation is behavior consistent with password guessing and service discovery. However, the records do not establish malicious intent, attacker identity or successful compromise. Operational context, log-source validation and additional authentication and network evidence are required before confirming an incident.

Lab cleanup

After completing the experiments and simulator activity, we restored the original security.log file from security.log.bak using Copy-Item with the -Force option. We then ran docker compose --profile simulation down to stop and remove the lab containers and shut down the simulation environment.

Overall conclusions

The controlled experiments confirmed that the detector behaves consistently with its rules. C1 demonstrated the effect of the minimum failure threshold, while C2 showed that failures are counted separately for each source IP. C3 confirmed that one new IP reaching the threshold generates a finding. C4 showed that an explicit scan notice generates an additional detection.

The simulator demonstrated how accumulated synthetic events can increase the number of findings. The local STIX data supplied technique names and tactics, making the results easier to interpret. However, enrichment does not strengthen the underlying evidence or confirm an attack. The main lesson is to trace every finding back to its log records, explain the rule that triggered it and keep observed facts separate from security interpretations.